

Assignment # 01

Course title:

Calculus and Analytical Geometry

Course code:

GESE-104

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Section:

AD

Table of Contents

Q1. Functions: 3
Q2. Types of Functions: 4
1. One-to-One Function 4
2. Many-to-One Function 5
3. Onto Function 6
4. Into Function 6
5. Bijective Function 7
6. Constant Function 7
7. Identity Function 8
8. Linear Function 8
Q3. Applications of Functions 9

Q1. Functions:

Definition:

A function is a mathematical relationship between two sets in which every element of the first set (domain) is associated with exactly one element of the second set (codomain).

A function is written as:

$f: A \rightarrow B$

where A is the domain and B is the codomain.

For example:

$f(x) = 2x + 3$

If $x = 2$

$f(2) = 2(2) + 3 = 7$

Thus, 2 is the input and 7 is the output.

Components of a Function

1. Domain:

The set of all possible input values of a function.

2. Codomain:

The set in which the output values are defined.

3. Range:

The actual output values obtained from the function.

Representation of Functions

Functions can be represented in different ways:

1. Ordered Pairs:

$f = \{(1,2), (2,4), (3,6)\}$

2.Table:

x	f(x)
1	2
2	4
3	6

3. Formula:

$f(x) = 2x$

4. Graph:

A function can also be represented by plotting its input and output values on a coordinate plane.

Example

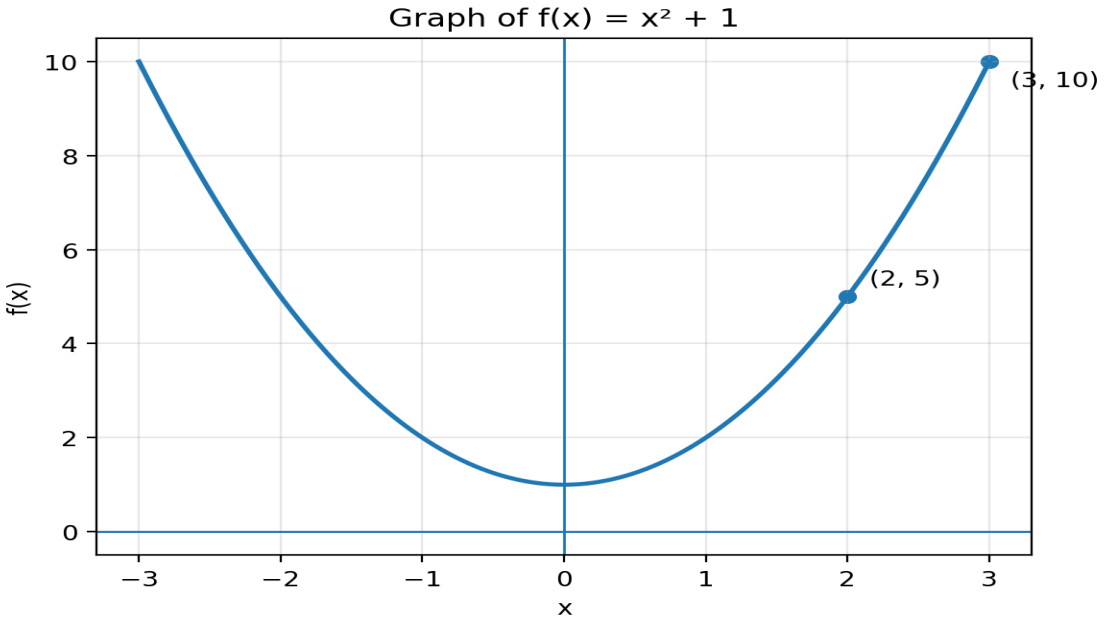
$f(x) = x^2 + 1$

For x = 2:

$f(2) = (2)^2 + 1 = 5$

For x = 3:

$f(3) = (3)^2 + 1 = 10$



Therefore, different input values produce corresponding output values.

Q2. Types of Functions:

Functions can be classified into different types according to the relationship between their input and output values.

1. One-to-One Function

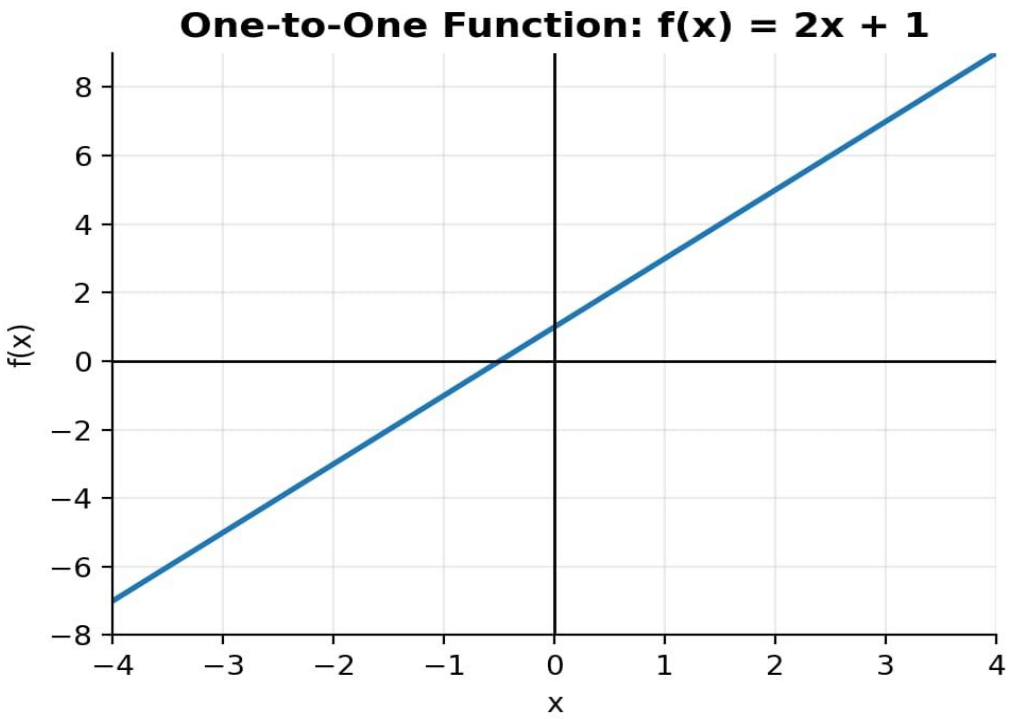
A function is one-to-one when different input values have different output values.

For example:

$f(x) = 2x + 1$

Here:

X	0	1	2	3
Y=2x+1	1	3	5	7



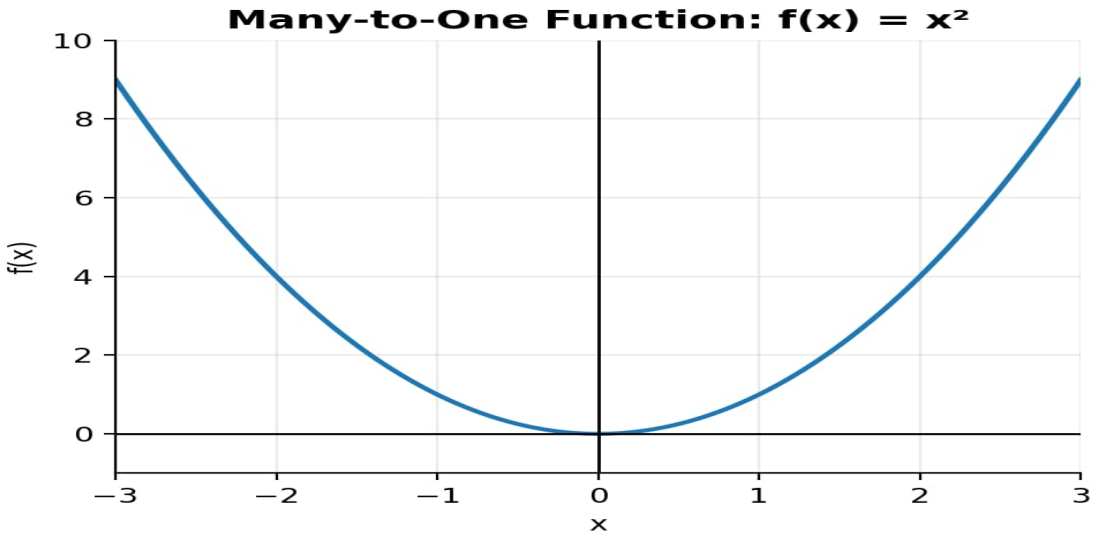
No two inputs have the same output.

2. Many-to-One Function

A function is many-to-one when two or more different inputs have the same output.

Example:f(x) = x^2

x	-2	-1	0	1	2
$y = x^2$	4	1	0	1	4



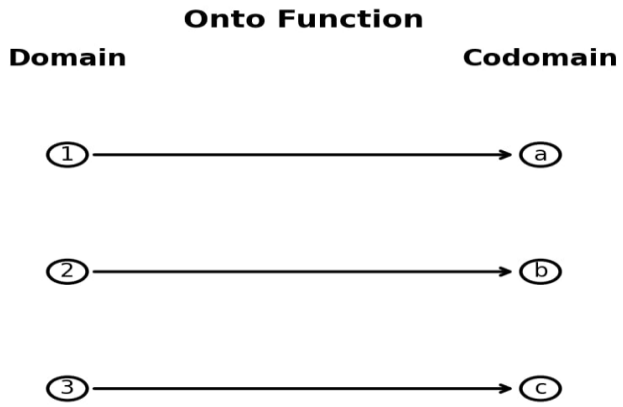
3. Onto Function

A function is called onto when every element of the codomain has at least one corresponding element in the domain.

For example, if:

x	1	2	3
f(x)	a	b	c

So:

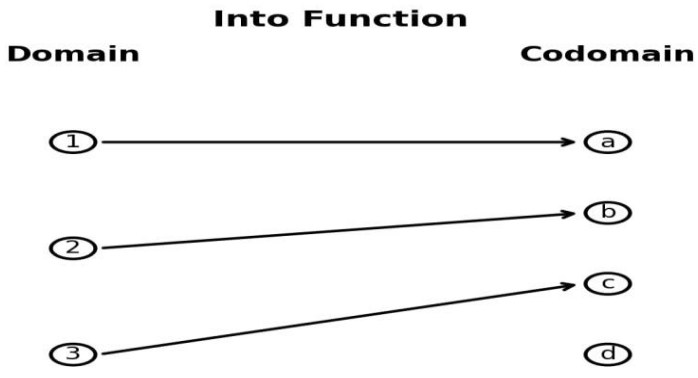


4. Into Function

A function is called into when at least one element of the codomain is not an output of any element of the domain.

For example:

X	**1**	**2**	**3**
f(x)	a	b	c



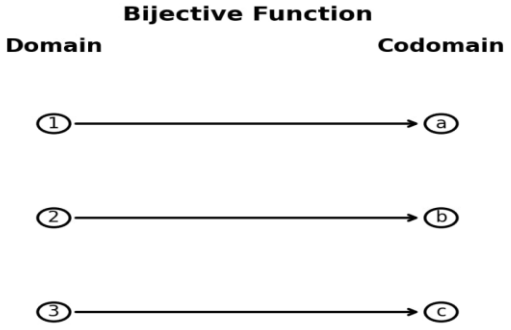
Here d is not used, so it is an into function.

5. Bijective Function

A function that is both one-to-one and onto is called a bijective function.

In a bijective function, every input has a unique output and every element of the codomain is used.

X	1	2	3
f(x)	a	b	c



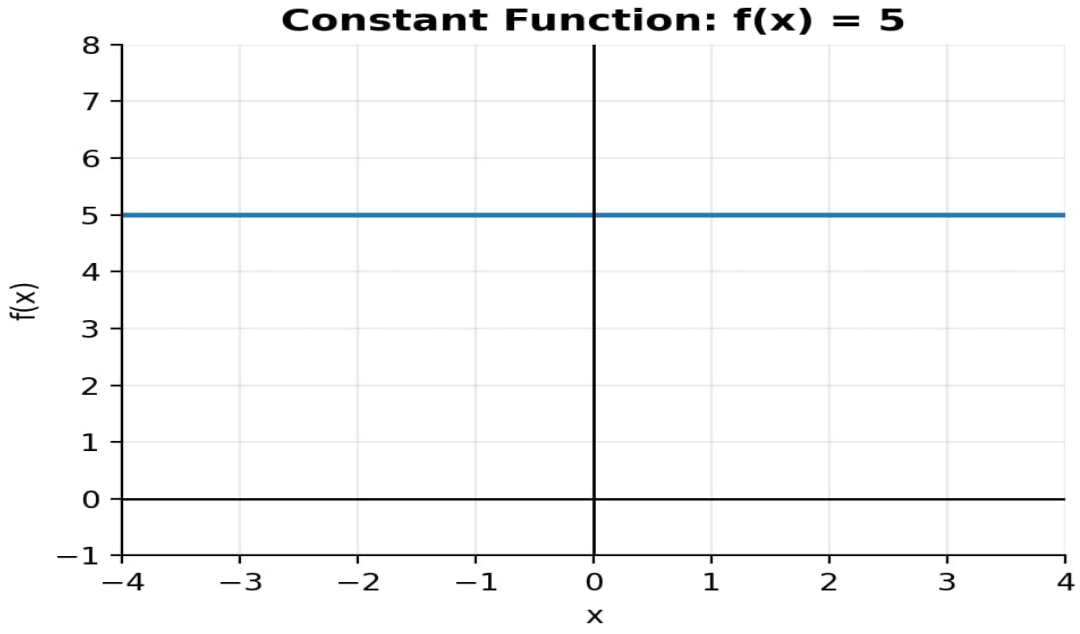
Every input has a unique output, and every element of the codomain is used.

6. Constant Function

A function in which every input produces the same output is called a constant function.

Example:f(x)=5

X	-2	-1	0	2
Y=5	5	5	5	5

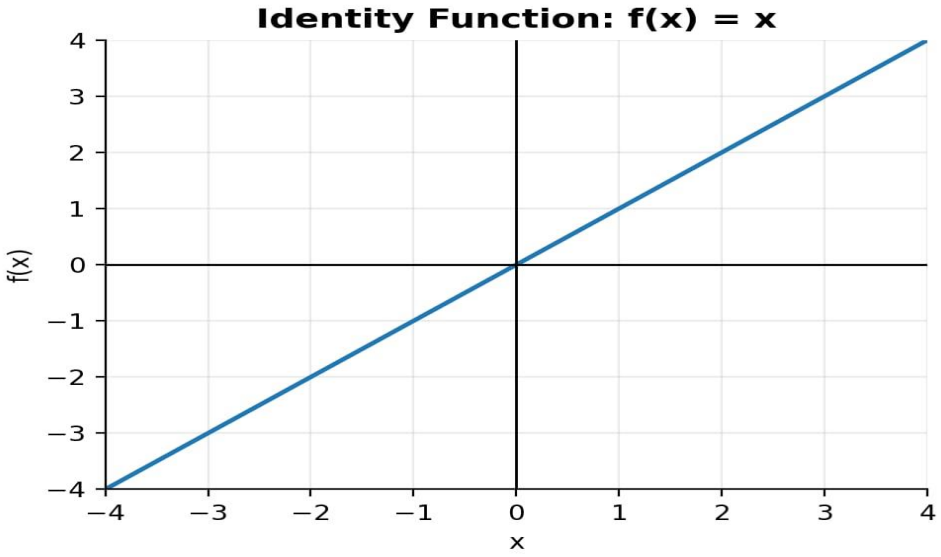


7. Identity Function

A function in which the output is equal to the input is called an identity function.

It is represented as,**f(x) = x**

X	**-2**	**-1**	**0**	**1**
y = x	-2	-1	0	1



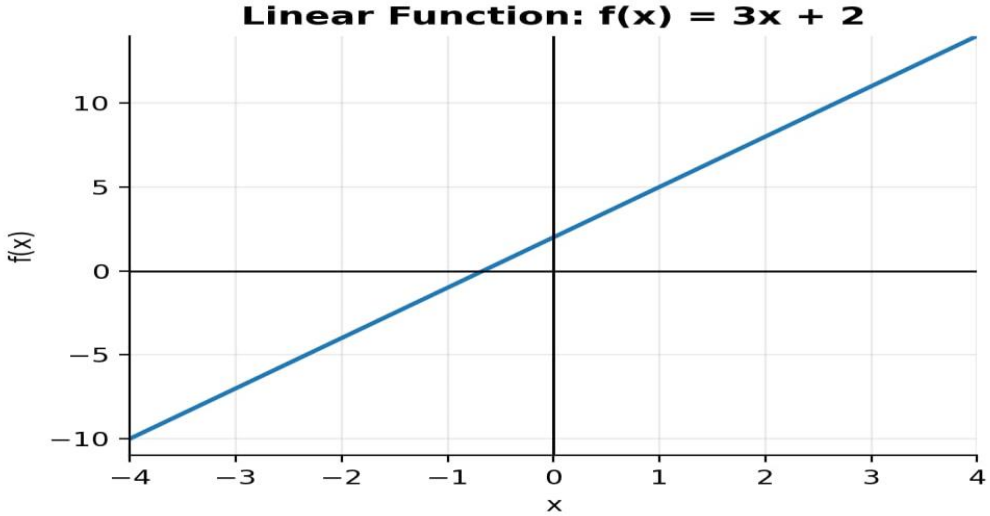
8. Linear Function

A function whose graph is a straight line is called a linear function.

General form:f(x) = mx + c

Example:f(x) = 3x + 2

x	**-2**	**-1**	**0**	**1**	**2**
y = 3x + 2	-4	-1	2	5	8



Q3. Applications of Functions

Functions are not only used in mathematics; they are also used to represent real-life relationships and solve practical problems.

1. Application in Physics:

Functions are used to describe the relationship between physical quantities.

For example, distance travelled at constant speed can be represented by:

$d(t) = vt$

where d is distance, v is velocity, and t is time.

If $v = 20$ m/s and $t = 5$ seconds:

$d = 20 \times 5 = 100$ m

2. Application in Computer Science:

Functions are an important part of programming. A programming function is a block of instructions designed to perform a particular task.

For example, a function can calculate the square of a number:

square(5) = 25

Functions make programs organized, reusable, and easier to understand.

3. Application in Economics:

Functions are used to represent economic relationships such as demand, supply, cost, revenue, and profit.

For example:

Profit = Revenue − Cost

If revenue and cost depend on the number of products sold, profit can be represented as a function of the number of products.

4. Application in Engineering:

Engineers use functions to model and analyze different systems.

For example, Ohm's Law is:

$V = IR$

where V is voltage, I is current, and R is resistance.

If resistance is constant, voltage can be represented as a function of current.

5. Application in Business:

Businesses use functions to calculate:

- Total cost
- Revenue
- Profit
- Production
- Sales
- Demand

For example, if the cost of producing products depends on the number of products produced, the total cost can be represented as a cost function.

6. Application in Daily Life:

Functions are also useful in everyday activities. Examples include:

- Calculating electricity bills
- Calculating travel time
- Calculating fuel consumption
- Calculating shopping discounts
- Calculating salaries
- Calculating mobile data usage

For example, a taxi fare can be represented as a function of the distance travelled.

THE END